

Yagnesh Jaidapuram

Machine Learning Engineer

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SUMMARY

Results-driven **Machine Learning Engineer** with 5+ years of experience designing, deploying, and optimizing predictive models and large-scale data pipelines across healthcare and enterprise environments. Skilled in building production-grade ML solutions that improve fraud detection, risk adjustment, and clinical analytics, achieving measurable business impact such as **94%+ model accuracy, 22% fraud detection improvement, and 40% faster data pipelines**. Proficient in **Python, TensorFlow, PyTorch, SQL, PySpark, and cloud platforms (AWS, Azure, GCP)** with hands-on expertise in **NLP, deep learning, MLOps, and model interpretability**. Adept at transforming terabyte-scale healthcare data into actionable insights, ensuring **HIPAA compliance**, and enabling real-time analytics for millions of members. Strong collaborator and mentor with proven ability to lead cross-functional teams, deliver executive-level presentations, and drive enterprise-wide adoption of ML and data management best practices.

EXPERIENCE

Machine Learning Engineer

Remote

Blue Cross Blue Shield

09/2024 - Present

- Designed, trained, and deployed predictive ML models on claims, clinical, and member engagement data to improve healthcare outcomes and reduce fraud risk.
- Designed and deployed over seven production-grade machine learning models, including predictive analytics, fraud detection, and NLP, achieving over 94% accuracy in claims processing and healthcare trend forecasting.
- Processed and analyzed over 10 million insurance claims and patient records using PySpark, SQL, and AWS, improving fraud detection by 22% and reducing false positives by 18%.
- Built and optimized automated data pipelines using Airflow, Kafka, and Spark, cutting model training time by 40% and enabling scalable ingestion of terabyte-scale healthcare data.
- Implemented deep learning techniques such as LSTMs, Transformers, and CNNs to improve predictive modeling for member risk scores, resulting in a 30% improvement in early detection rates for high-risk patients.
- Deployed models on cloud platforms including AWS SageMaker, Azure ML, and Databricks with CI/CD integration, reducing the deployment cycle from weeks to days.
- Developed explainable AI frameworks using SHAP and LIME to enhance model interpretability, ensuring compliance with HIPAA and federal audit requirements.
- Partnered with data governance and compliance teams to establish enterprise-level ML best practices, improving data reliability and compliance audit scores by 25%.
- Designed real-time monitoring dashboards with Tableau, Power BI, and Grafana to track ML performance and business KPIs, reducing manual reporting by 50%.
- Mentored two junior ML engineers and interns by conducting code reviews, architecture walkthroughs, and model performance optimization sessions.
- Presented predictive analytics solutions to executives and healthcare leaders, directly influencing strategies that impacted over one million members nationwide.

Machine Learning Intern

Austin, TX

McKesson

07/2023 - 08/2024

- Collaborated with stakeholders across verticals to implement Data Management & Global Records Management policy standards.
- Designed, trained, and deployed 10+ machine learning models (classification, regression, and NLP) using Python, TensorFlow, and Scikit-learn, achieving up to 95% accuracy in predictive healthcare analytics.
- Engineered data pipelines to process and cleanse 5M+ structured and unstructured healthcare records, improving model training efficiency and reducing data errors by 28%.
- Implemented feature engineering and dimensionality reduction techniques (PCA, embeddings), boosting model precision and recall by 20%.
- Automated ETL workflows with PySpark and SQL, cutting data ingestion time from 4 hours to under 45 minutes.
- Deployed production-ready ML models on AWS SageMaker & Azure ML, reducing prediction latency by 35% and enabling near real-time analytics for clinical operations.
- Developed interactive dashboards (Power BI & Tableau) showcasing predictive insights, reducing executive reporting turnaround by 40%.
- Partnered with cross-functional teams (10+ data engineers, analysts, and business stakeholders) to translate complex healthcare data into actionable insights supporting patient care and cost-optimization strategies.
- Conducted rigorous model validation using A/B testing and advanced metrics (AUC-ROC, F1-score), ensuring compliance with HIPAA regulations and delivering 98% data accuracy in outputs.
- Optimized hyperparameters and experimented with ensemble methods, which improved model robustness and delivered a 15% uplift in predictive performance over baseline models.
- Authored comprehensive documentation covering workflows, model interpretability (SHAP, LIME), and deployment steps, enabling seamless handoff to full-time ML engineers.
- Presented findings to senior leadership in 5+ strategic meetings, influencing decisions on digital health solutions that impacted 10K+ patients nationwide.

EXPERIENCE

Data Management Consultant

India

Cts

08/2019 - 11/2021

- Designed and implemented data quality control rules for enterprise data warehouse platforms
- Wrote Python scripts to automate data validation and reconciliation processes
- Managed ETL processes and supported CRM/ERP data flows across departments
- Built dashboards and visualizations using MicroStrategy to report data quality KPIs to leadership
- Conducted root cause analysis on data incidents and worked with engineering teams to define long-term solutions
- Designed and deployed a multi-cloud architecture across GCP and Azure for a global banking client, enabling 99.99% uptime for regulatory workloads and reducing latency by 30% for cross-regional transaction processing
- Developed an internal self-service developer platform using Backstage, reducing developer onboarding time by 50% and increasing CI/CD pipeline adoption by 40%, accelerating delivery of digital banking features
- Managed enterprise-scale Kubernetes clusters with over 1,000+ pods, implemented automated autoscaling and enforced pod security policies, improving infrastructure efficiency by 35% and meeting zero-trust and PCI-DSS compliance requirements
- Standardized over 50 Terraform modules across dev, QA, and prod environments, reducing configuration drift incidents by 70%, and improving audit readiness and compliance tracking for internal IT governance teams

Data Management Consultant Intern

Hyderābād, India

Trigent

08/2018 - 02/2019

- Assisted in gathering and analyzing client requirements, contributing to five or more solution design documents that aligned business needs with technical specifications.
- Processed and validated over 200,000 data records using SQL queries and Excel models, improving reporting accuracy by 18%.
- Supported the development of ETL workflows (SSIS/SSRS), reducing data load time by 15% and ensuring reliable integration of large datasets.
- Designed and delivered five interactive dashboards in Power BI, increasing visibility into business KPIs for more than ten client stakeholders.
- Conducted more than 25 data quality checks and profiling tasks, detecting inconsistencies that improved compliance and reduced data errors by 12%.
- Documented business requirements, user stories, and process workflows, enhancing communication between client-facing and technical teams.
- Recommended two process improvements during an internship, streamlining manual reporting tasks and saving approximately five hours per week for the project team.

EDUCATION

Masters in Computer information systems

TN,US

Christian Brother's University

01/2022 - 05/2023

Bachelor's in computer science and engineering

Hyderabad,India

Sphoorthy Engineering College

08/2015 - 05/2019

SKILLS

Programming & ML Frameworks:

Python, R, PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, XGBoost, LightGBM

Data Engineering & Big Data:

SQL, PySpark, Apache Kafka, Airflow, Databricks, ETL Pipelines, Data Warehousing (Snowflake, Redshift)

Cloud & Deployment:

AWS (SageMaker, S3, EC2, Lambda), Azure ML, GCP AI Platform, Docker, Kubernetes, CD/CI GitHub Actions, Jenkins

Machine Learning & AI:

Supervised & Unsupervised Learning, Deep Learning (CNNs, RNNs, LSTMs, Transformers), NLP, Generative AI, Reinforcement Learning, Recommendation Systems, Time Series Forecasting, Computer Vision

Model Optimization & Monitoring:

Feature Engineering, Hyperparameter Tuning, Model Interpretability (SHAP, LIME), A/B Testing, MLOps, Model Drift Detection

Data Visualization & Analytics: Tableau, Power BI, Grafana, Matplotlib, Seaborn, Excel (Advanced), Interactive Dashboards

Healthcare Domain Knowledge:

Claims Processing, Risk Adjustment Models, Fraud Detection, HIPAA Compliance, Clinical Data Analytics

Soft Skills & Collaboration:

Agile/Scrum, Cross-functional Collaboration, Technical Documentation, Stakeholder Communication, Mentorship